

R2 Score value details across Algorithms as given below:

1. Multiple Regression:

Ans: R2 Value = 0.935

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03-Hope-AI---Machine-Learning-Assignments / 02 Multiple Linear Regression Assignment.ipynb

Preview Code Blame 2866 lines (2866 loc) · 101 KB

[ 88040.6718287 ],
[110949.99405525],
[ 90419.1897851 ],
[128020.46250064]])

In [12]: from sklearn.metrics import r2_score
r_score = r2_score(y_test,y_pred)
r_score

Out[12]: 0.9358680970046241
```

2. SVM – Support Vector Machine:

3. Support Vector Machine					
Serial#	Hyper Parameter	Linear (r value)	rbf (Non-Linear Value)	Poly (r value)	Sigmoid (r value)
1	C10	-2.437	-0.055	0.025	-0.057
2	C100	-357.079	-0.030	0.465	-0.058
3	C500	Too much time to run	0.050	0.620	-0.064
4	C1000	Too much time to run	0.160	0.640	-0.070
5	C2000	Too much time to run	0.288	0.671	-0.084
6	C3000	Too much time to run	0.395	0.690	-0.098

3. Decision Tree:

4. Decision Tree				
Serial#	Criterion	Splitters	Max Features	R2_Value
1	<i>squared_error</i>	<i>best</i>	<i>int (None)</i>	0.925
2	<i>squared_error</i>	<i>random</i>	<i>int (None)</i>	0.828
5	<i>squared_error</i>	<i>best</i>	<i>sqrt</i>	0.866
6	<i>squared_error</i>	<i>random</i>	<i>sqrt</i>	0.795
7	<i>squared_error</i>	<i>best</i>	<i>log2</i>	0.741
8	<i>squared_error</i>	<i>random</i>	<i>log2</i>	0.779
9	<i>absolute_error</i>	<i>best</i>	<i>int (None)</i>	0.949
10	<i>absolute_error</i>	<i>random</i>	<i>int (None)</i>	0.765
11	<i>absolute_error</i>	<i>best</i>	<i>sqrt</i>	0.764
12	<i>absolute_error</i>	<i>random</i>	<i>sqrt</i>	0.840
13	<i>absolute_error</i>	<i>best</i>	<i>log2</i>	0.085
14	<i>absolute_error</i>	<i>random</i>	<i>log2</i>	0.527
15	<i>poisson</i>	<i>best</i>	<i>int (None)</i>	0.930
16	<i>poisson</i>	<i>random</i>	<i>int (None)</i>	0.832
17	<i>poisson</i>	<i>best</i>	<i>sqrt</i>	0.488
18	<i>poisson</i>	<i>random</i>	<i>sqrt</i>	-0.545
19	<i>poisson</i>	<i>best</i>	<i>log2</i>	0.666
20	<i>poisson</i>	<i>random</i>	<i>log2</i>	0.741